IOWA STATE UNIVERSITY

LAB 1 - BEAM DESIGN REPORT

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Aer E 4260 Design of Aerospace Structures

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1

Introduction

In Lab 1, we were tasked with designing the front spar of an aircraft wing, modeled as an I-beam. The primary design constraint was the maximum allowable deflection, with the length of the wing, beam height, material properties of 2024 aluminum, and load distribution provided. The loading was simplified as a triangular distribution, rather than the more complex elliptical distribution typically encountered in aerodynamic analysis. Given this information, we were responsible for determining the thicknesses of the flange (tf) and web (tw) sections, as well as the depth (d) of the I-beam. These dimensions had to satisfy both the given design constraints and ensure an appropriate factor of safety for the spar under the applied loads.

2

Beam Geometry and Material Properties

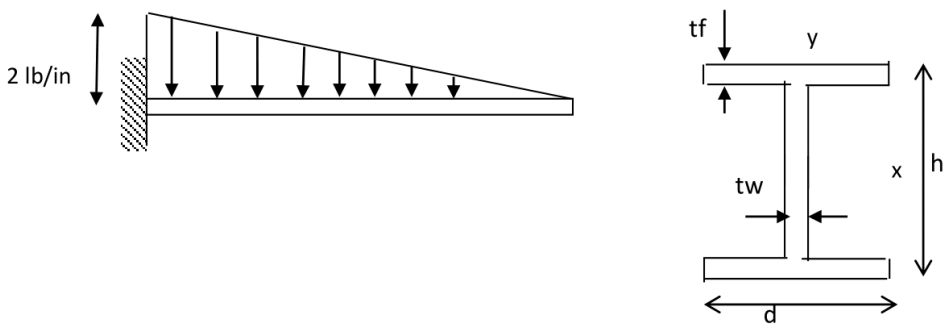


Figure 2.1: *Frontal and side view of the beam, showing the distributed load, and dimensions of the I-beam (tf, tw, d, and h).*

Table 3.2.3.0(e_1). Design Mechanical and Physical Properties of Clad 2024 Aluminum Alloy Sheet and Plate

Specification	AMS-QQ-A-250/5																			
Form	Flat sheet and plate																			
Temper	T3								T351											
Thickness, in.	0.008-0.009		0.010-0.062		0.063-0.128		0.129-0.249		0.250-0.499		0.500-1.000[a]		1.001-1.500[a]		1.501-2.000[a]		2.001-3.000[a]		3.001-4.000[a]	
Basis	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B
Mechanical Properties:																				
F_{tu}, ksi:																				
L	59	60	60	61	62	63	63	64	62	64	61	63	60	62	60	62	58	60	55	57
LT	58	59	59	60	61	62	62	63	62	64	61	63	60	62	60	62	58	60	55	57
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	52[b]	54[b]	49[b]	51[b]
F_{ty}, ksi:																				
L	44	45	44	45	45	47	45	47	46	48	45	48	45	48	45	47	44	46	39	41
LT	39	40	39	40	40	42	40	42	40	42	40	42	40	42	40	42	40	42	39	41
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	38[b]	40[b]	38[b]	39[b]
F_{cy}, ksi:																				
L	36	37	36	37	37	39	37	39	37	39	37	39	37	39	36	38	35	37	33	35
LT	42	43	42	43	43	45	43	45	43	45	42	45	42	44	42	44	41	43	39	41
ST	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	46	48	44	47
F_{sy}, ksi	37	37	37	38	38	39	39	40	37	38	36	37	35	37	35	37	34	35	32	34
F_{bru}[c], ksi:																				
(e/D = 1.5)	96	97	97	99	101	102	102	104	94	97	92	95	91	94	91	94	88	91	83	86
(e/D = 2.0)	119	121	121	123	125	127	127	129	115	119	113	117	111	115	111	115	107	111	102	106
F_{bry}[c], ksi:																				
(e/D = 1.5)	68	70	68	70	70	73	70	73	69	72	69	72	69	72	69	72	69	72	67	70
(e/D = 2.0)	82	84	82	84	84	88	84	88	82	86	82	86	82	86	82	86	82	86	80	84
e, percent (S-basis):																				
LT	10	...	[4]	...	15	...	15	...	12	...	8	...	7	...	6	...	4	...	4	...
E, 10^3 ksi:																				
Primary	10.5								10.7											
Secondary	9.5				10.0				10.2											
E_c, 10^3 ksi:																				
Primary	10.7								10.9											
Secondary	9.7				10.2				10.4											
G, 10^3 ksi	...																			
μ	0.33																			
Physical Properties:																				
ω, lb/in.^{3}	0.100																			
C, K, and α	...																			

a These values, except in the ST direction, have been adjusted to represent the average properties across the whole section, including the 2-½ percent nominal cladding thickness.
b Caution: This specific alloy, temper, and product form exhibits poor stress-corrosion cracking resistance in this grain direction. It corresponds to an SCC resistance rating of D, as indicated in Table 3.1.2.3.1(a).
c Bearing values are "dry pin" values per Section 1.4.7.1. See Table 3.1.2.1.1.
d See Table 3.2.3.0(f).

Table 2.1: *Design mechanical and physical properties of clad 2024 aluminum alloy sheet and plate.*

3

Methodology

The first step in the design process was to address the primary constraint: maximum allowable deflection of 15 inches. Since we are working with a wing, the maximum deflection occurs at the wing tip. Thus, deflection at the wing tip cannot exceed 15 inches. Equation (3.1) relates deflection to the load, length, Young's modulus, and moment of inertia about the center x-axis:

$$\delta = \frac{wL^4}{30EI} \tag{3.1}$$

Rearranging (3.1), we derive an expression for the minimum required moment of inertia about the center x-axis:

$$I_{\text{min}} = \frac{wL^4}{30E\delta_{\text{max}}} \tag{3.2}$$

From (3.2), we can calculate the minimum required moment of inertia based on the maximum allowable deflection, known length, Young's modulus, and maximum load.

The design mechanical and physical properties of clad 2024 aluminum alloy sheet and plate are listed in Table 2.1. The chosen Young's modulus value was for temper T351 in compression, $E_c = 10.9$ ksi, as it is the most limiting for designing I_{min}. Considering the problem assumption of a solid wing, the effective Young's modulus is doubled:

$$E_{\text{eff}} = 2E_c = 2 \times 10900 \text{ ksi} = 21800 \text{ ksi} \tag{3.3}$$

By plugging in the values and converting units, we calculate the minimum required moment of inertia about the x-axis:

$$I_{\text{min}} = \frac{2\,\text{lb/in} \times (50\,\text{ft})^4}{30 \times (21800\,\text{ksi}) \times 15\,\text{in}} = 26.422\,\text{in}^4 \tag{3.4}$$

For the initial design, it was assumed that $t_w = t_f = t$. Using the following moment of inertia equation, we can calculate t_{min} based on I_{min}:

$$I = 2\left[\frac{1}{12}dt_f^3 + dt_f\left(\frac{h}{2}\right)^2\right] + \frac{1}{12}t_w h^3 \tag{3.5}$$

Since t_f^3 is much smaller than 1, the first term of the equation can be approximated as 0. Additionally, for the first design, we estimate $d = 4\,\text{in}$. Substituting values:

$$26.422\,\text{in}^4 = 2 \times 4\,\text{in} \times t_{\text{min}} \times \left(\frac{9\,\text{in}}{2}\right)^2 + \frac{1}{12} t_{\text{min}} \times (9\,\text{in})^3 \tag{3.6}$$

This gives $t_{\text{min}} = 0.119\,\text{in}$.

For easier manufacturing and to exceed the minimum limit, we select $t_f = t_w = 0.2\,\text{in}$. With the three initial design variables (t_f, t_w, and d), we calculate the moment of inertia I, maximum axial stress σ_{max}, and shear stress τ as follows:

From (3.5):

$$I = 2\left[\frac{1}{12} \times 4\,\text{in} \times (0.2\,\text{in})^3 + 4\,\text{in} \times 0.2\,\text{in} \times \left(\frac{9\,\text{in}}{2}\right)^2\right] + \frac{1}{12} \times 0.2\,\text{in} \times (9\,\text{in})^3 = 44.555\,\text{in}^4 \tag{3.7}$$

Moment:

$$M = \frac{wL^2}{6} = \frac{2\,\text{lb/in} \times (50\,\text{ft})^2}{6} = 1.2 \times 10^5\,\text{lb} \cdot \text{in} \tag{3.8}$$

Maximum axial stress:

$$\sigma_{\text{max}} = \frac{My_{\text{max}}}{I} = \frac{1.2 \times 10^5\,\text{lb} \cdot \text{in} \times 4.5\,\text{in}}{44.555\,\text{in}^4} = 12.120\,\text{ksi} \tag{3.9}$$

Moment of area:

$$Q = dt_f \frac{h}{2} + t_w \frac{h^2}{8} = 5.625\,\text{in}^2 \tag{3.10}$$

Shear force:

$$V = \frac{wL}{2} = \frac{2\,\text{lb/in} \times 50\,\text{ft}}{2} = 600\,\text{lb} \tag{3.11}$$

Shear stress:

$$\tau = \frac{VQ}{It_w} = \frac{600\,\text{lb} \times 5.625\,\text{in}^2}{44.555\,\text{in}^4 \times 0.2\,\text{in}} = 0.379\,\text{ksi} \tag{3.12}$$

Deflection:

$$\delta = \frac{wL^4}{30EI} = \frac{2\,\text{lb/in} \times (50\,\text{ft})^4}{30 \times (21000\,\text{ksi}) \times 44.555\,\text{in}^4} = 9.23\,\text{in} \tag{3.13}$$

Thus, the maximum deflection of 15 inches is not exceeded.

Given that $t_w = t_f = 0.2\,\text{in}$, we select temper T3 for the 2024 aluminum based on Table 2.1. The yield compressive axial stress (most limiting), yield shear stress and maximum allowable deflection are used to calculate the factors of safety:

$$FS_{\sigma_{\text{yield}}} = \frac{\sigma_{\text{yield}}}{\sigma_{\text{max}}} = \frac{37\,\text{ksi}}{12.120\,\text{ksi}} = 3.05 \tag{3.14}$$

$$FS_{\tau_{\text{yield}}} = \frac{\tau_{\text{yield}}}{\tau} = \frac{39\,\text{ksi}}{0.379\,\text{ksi}} = 102 \tag{3.15}$$

$$FS_{\delta} = \frac{\delta_{\text{max}}}{\delta} = \frac{15\,\text{in}}{9.23\,\text{in}} = 1.63 \tag{3.16}$$

4

Conclusion

The front spar of the aircraft wing was successfully designed to meet the maximum allowable deflection constraint of 15 inches. The final design used 2024-T3 aluminum with thicknesses $t_f = 0.2\,\text{in}$, $t_w = 0.2\,\text{in}$, and depth $d = 4\,\text{in}$. The calculated moment of inertia, maximum axial stress, and shear stress were within acceptable limits, and the factor of safety was calculated to be 3.05 for axial stress and 102 for shear stress. This ensures the design meets both the strength and deflection requirements while providing a safe and reliable solution.

However, the design could be further optimized by adjusting the dimensions or material properties to reduce weight or improve performance while still maintaining the required safety margins. Additional refinements and iterations could lead to a more efficient structural design.

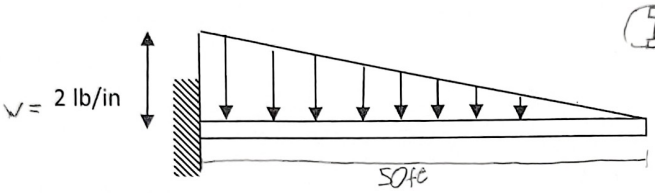
A

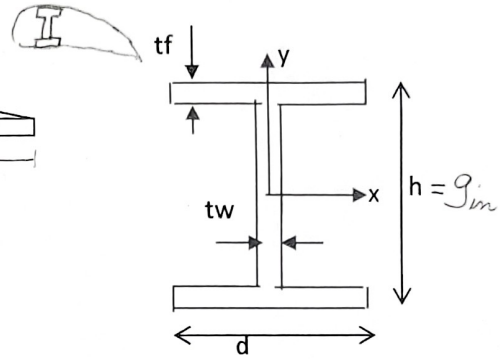
Appendix A

A.1 Rough Work

AerE 526/426
Lab Exercise-1

You have to design the front spar of an aircraft. The length of the wing has been determined and should be 50 ft. The height, h, of the beam from the airfoil design is 9 inches. The maximum deflection should not exceed 15 inches. The aerodynamic studies have calculated the loading of the beam, and the load distribution is given below. (In reality it should have been an elliptical distribution but to keep things simple I have made it into a triangular distribution).
The material properties of 2024 aluminum are given in the enclosed chart. After you are done with your design of the front spar, calculate the correct Factor of safety for your design. Since we have assumed the wing to be solid in place of a hollow section, the effective modulus of the wing is double of the modulus of aluminum. Material properties are given at the back of the page. Can you improve your solution, maybe, one more step of approximation to get a better solution?





Just to help you, the equations you will use:

$$\delta = \frac{wL^4}{30EI}$$

$Moment\ M = \frac{wL^2}{6}$ $Shear\ V = \frac{wL}{2}$, w is max given =2lb/in

$$Axial\ Stress\ \max\sigma = \frac{My_{max}}{I}$$

$$Shear\ stress\ \tau = \frac{V * Q}{I * t_w}$$

$$momet\ of\ inertia\ about\ center\ x\ axis\ I = 2\left[\frac{1}{12}d * t_f^3 + d * t_f * \left(\frac{h}{2}\right)^2\right] + \frac{1}{12}t_w(h)^3$$

$$moment\ of\ area\ Q = d * t_f * \frac{h}{2} + t_w\frac{h^2}{8}$$

$$G = \frac{E}{2(1+\mu)}$$

Instructions for submission:

1. You are allowed to work together in groups.
2. The report will be individual.
3. Work is due 1 week after the lab.
4. Rewrite your work in a report form and state your methodology, approximations and assumptions.
5. Append the rough work you did in the class.

In the following: Ftu: Tension ultimate, Fty: Tension yield, Fcy: Compression yield, Fsy: Shear Yield, Fbru: Bearing ultimate, Fbry: Bearing yield, L: Long (roll) direction, LT: Transverse to roll, ST: Thickness direction. E: Modulus Tension, Ec: Modulus Compression, G: Shear modulus, μ: Poisson's ratio.
In this chart A refers to 99% reliable values and B means 90% reliable values.

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ST	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	46	48	44	47
Fsy, ksi	37	37	37	38	38	39	39	40	37	38	36	37	35	37	35	37	34	35	32	34
Fbru[c], ksi:																				
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e, percent (S-basis):																				
LT	10	...	4	...	15	...	15	...	12	...	8	...	7	...	6	...	4	...	4	...
E, 10^3 ksi:																				
Primary	10.5								10.7											
Secondary	9.5				10.0				10.2											
Ec, 10^3 ksi:																				
Primary	10.7								10.9											
Secondary	9.7				10.2				10.4											
G, 10^3 ksi																				
μ	0.33																			
Physical Properties:																				
ω, lb/in^3	0.100																			
C, *K*, and α																				

a These values, except in the ST direction, have been adjusted to represent the average properties across the whole section, including the 2-½ percent nominal cladding thickness.
b Caution: This specific alloy, temper, and product form exhibits poor stress-corrosion cracking resistance in this grain direction. It corresponds to an SCC resistance rating of D, as indicated in Table 3.1.2.3.1(a).
c Bearing values are "dry pin" values per Section 1.4.7.1. See Table 3.1.2.1.1.
d See Table 3.2.3.0(f).

AERE 421 - Lab Assignment 1

. $\delta_{max} = 15\ in$. $\delta = \frac{wL^4}{30 \cdot E \cdot I} \Rightarrow I = \frac{wL^4}{30E\delta}$

. $E = 10.9 \times 10^3\ ksi$ (assuming E mean between T and C from T31)

. Total load $= \frac{w \cdot L}{2} = \frac{2(50 \cdot 12)}{2} = 600\ lb$. $E_{eff} = 2E$

$$I_{min} = \frac{(2\ lb/in)(50 \cdot 12\ in)^4}{30(21.8 \times 10^6\ psi)(15\ in)} = 26.422\ in^4$$

$$M_{max} = \frac{(2\ lb/in)(50 \cdot 12\ in)^2}{6} = 1.2 \times 10^5\ lb \cdot in$$

$$V_{max} = \frac{(2\ lb/in)(50 \cdot 12\ in)}{2} = 600\ lb$$

$$\sigma_{max} = \frac{(1.2 \times 10^5\ lb \cdot in)(4.5\ in)}{26.667\ in^4} = 20.25\ ksi$$

(1) $I = \frac{M y_{max}}{\sigma} = \frac{(1.2 \times 10^5\ lb \cdot in)(4.5\ in)}{(46 \times 10^3\ psi)} = 11.7\ in^4 < I_{min}$

. Guess $d = 4\ in$

$$I = 2\left(\frac{d\,t_f^3}{12} + d \cdot t_f \cdot \left(\frac{h}{2}\right)^2\right) + \frac{t_w (h)^3}{12}$$

small small

$$I_{min} < \frac{d\,t_f h^2}{2} \Rightarrow t_f > \frac{2 I_{min}}{d\,h^2} \Rightarrow t_f > \frac{2(26.422\ in^4)}{(4\ in)(9\ in)^2}$$

Assuming $t_f = t_w$

$t_f > 0.163\ in \Rightarrow t_f = 0.2\ in$ $\qquad t_w = 0.2\ in$

$Q = d \cdot t_f \cdot \frac{h}{2} + t_w \frac{h^2}{8} = 5.625\ in^3$ $\qquad I = 44.555\ in^4$

$\tau = \frac{V Q}{I t_w} = 378.7\ psi$ $\qquad \sigma = \frac{M y_{max}}{I} = \frac{(1.2 \times 10^5\ lb \cdot in)(4.5\ in)}{(44.555\ in^4)}$

2024 Aluminum T351 : Ductile material

$\sigma = 12.120\ ksi$

$FS_\sigma = \frac{\sigma_Y}{\sigma} = \frac{46\ ksi}{12.120\ ksi} = 3.80$

$FS_\tau = \frac{\tau_y}{\tau} = \frac{37\ ksi}{0.379\ ksi} = 97.6$